CUDA & GPU - OVERVIEW

Example: AMD EPYC 9754 & NVIDIA H100 SXM

NVIDIA CUDA

- Compute kernel as C program
- Explicit data- and thread-level parallelism
- Computing, not graphics processing
- Host communication

Memory hierarchy

- Host memory
- GPU (device) memory
- GPU on-chip memory (later)

More HW details exposed

- Use of pointers
- Load/store architecture
- Barrier synchronization of thread blocks

